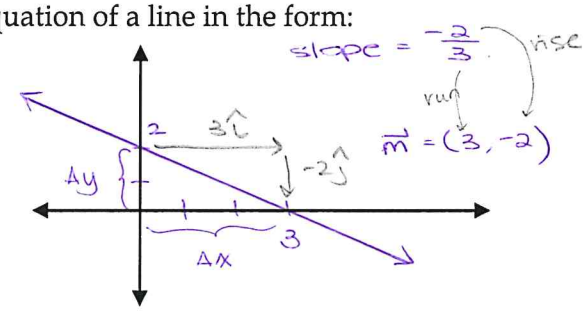
Vector and Parametric Equations of a Line in R²

Recall:

To determine an equation of a line in R², we would write the equation of a line in the form:

$y = mx + b$



To determine an equation of a line, we need:

1. slope ⇒ direction
2. point ⇒ location

1. Direction

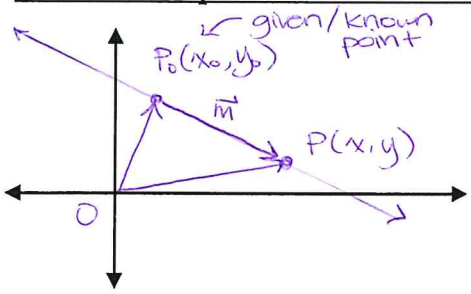
To use vectors, we first define the DIRECTION VECTOR for the line, $\vec{m}$.

A line in R² with the direction vector $\vec{m} = (a, b)$ has a slope of $\frac{b}{a}$.

2. Location

A point on the line has an associated POSITION VECTOR, $\overrightarrow{OP}$. Since a line is a set of points, we can consider that line to be a set of position vectors whose endpoints are on the line.

The Vector Equation of a line in R²



$\overrightarrow{OP} = \overrightarrow{OP_0} + \overrightarrow{P_0P}$, $\overrightarrow{P_0P} \not\parallel \vec{m}$ (collinear)

$\therefore \overrightarrow{P_0P} = t\vec{m}$, $t \in \mathbb{R}$.

scalar multiple

$\therefore \overrightarrow{OP} = \overrightarrow{OP_0} + t\vec{m}$

Let $\overrightarrow{OP} = \vec{r}$

$\overrightarrow{OP_0} = \vec{r_0}$

Vector Equation of a line in R²

$$\vec{r} = \vec{r}_0 + t\vec{m}, \; t \in \Re$$

in component form:

$(x, y) = (x_0, y_0) + t(a, b)$, $t \in \mathbb{R}$.

Parametric Equations of a line in R²

$(x, y) = (x_0, y_0) + t(a, b), \; t \in \Re$

$= (x_0, y_0) + (ta, tb)$

$= (x_0 + ta, \; y_0 + tb)$

$\therefore \; x = x_0 + ta$

$y = y_0 + ta$

, $t \in \mathbb{R}$

Example 1: Given a point $(2,3)$ with a direction vector $\vec{m} = (3,-1)$,

direction vector can be any scalar multiples

a) Determine the vector equation of this line.

$\vec{r} = \vec{r_0} + t\vec{m}$, $t \in \mathbb{R}$

$= (2,3) + t(3,-1)$, $t \in \mathbb{R}$.

OR $= (2,3) + s(6,-2)$, $s \in \mathbb{R}$

b) Determine the parametric equation of this line.

$x = 2 + 3t$

$y = 3 - t$, $t \in \mathbb{R}$.

Note: could write a different equation

$\vec{r} = (11,0) + t(3,-1)$, $t \in \mathbb{R}$

c) Determine the point on this line that corresponds with the parameter $t=3$.

using vector notation.

$\vec{r} = (2,3) + 3(3,-1)$

$= (2,3) + (9,-3)$

$\vec{r} = (11,0)$

using parametric

$x = 2 + 3(3)$ $y = 3 - (3)$

$x = 11$ $y = 0$

$\therefore$ point $(11,0)$

d) Determine if the point (1,-8) is on the line.

solve for parameter t:

$x = 2 + 3t$

$1 = 2 + 3t$

$-1 = 3t$

$\frac{-1}{3} = t$

$y = 3 - t$

$-8 = 3 - t$

$-11 = -t$

$11 = t$

$\therefore$ No! Not a single t-value.

Example 2: Determine a vector equation for the line that is perpendicular to the line $\vec{r} = (1,3) + s(-1,2)$ and passes through the point $(0,5)$.

Need a point

$(0,5)$

Need a direction vector

$\perp$ to $(-1,2)$

Let $\vec{m} = (a,b)$

$(a,b) \cdot (-1,2) = 0$

$-a + 2b = 0$

$a = 2b$.

let $b = 1$

$a = 2$.

$\therefore \vec{m} = (2,1)$

OR think negative reciprocals!

$m = \frac{-2}{1}$

$m_{\perp} = \frac{1}{2}$.

$\therefore \vec{r} = (0,5) + t(2,1)$, $t \in \mathbb{R}$